Automorphism ideas: implementation and experiment plan

2026-10-21 | Planning report | No experiments run

Repository fit and prior work

Reviewed against cryptanalysis:

- Base commit: `df69c0c3c35d6c30ad7f9b96e43c91b2be90b89e`.
- Base branch: `claude/cryptanalysis-repo-setup-p59kek`.

The candidate catalog already separates usable points from folded columns: its N13 profile records fb6 with two columns, and its N131 normal-basis profiles record fb3668 with 14 columns. These are existing catalog facts, not new measurements here. DGM-01 therefore tests an incremental change against that folded baseline; an unfurled arm is a correctness control.

PR 13 covers automorphism rho and PR 64 covers prime-field candidate distinctions. PR 416 and PR 442 are overlap pointers for orbit reducers and exponent transport. A PR pointer does not assert that its proposed implementation is merged or validated.

Proposed integration boundaries

The following existing files are inspection starting points, not claims that all necessary APIs already exist. Implementation should first trace current call sites and reuse their arithmetic and evidence formats.

Workstream	Starting points	New evidence required before promotion
DGM-01	`src/indexcalc.c`, `src/linalg.c`, candidate catalog	Stable point-set digest, orbit transport, current-folded ablation, rank and known-answer replay
DGM-02	`src/curve.c`, curve and route records	Exact map and inverse, subgroup scalar action, native certificate
DGM-03	`src/rho.c`, coefficient/restart handling	Quotient consistency, cycle escape, useful collisions and matched recovery
DGM-04	`src/curve.c`, `src/volcano.c`	Exact standard-curve records and native trace/discriminant exclusion check
DGM-05	New literature/design note before code	Full-paper review, explicit input assumptions and a non-oracular worked algorithm

Use the native C backend here and the companion crypto repository's native Rust certificate/design work where appropriate. No Python cryptanalytic harness is proposed. CPU toy correctness controls do not become IC performance claims; use the required GPU path when a later performance experiment is ready.

Select exact registered curve and subgroup IDs, full factor-base point digests, code pins and seed streams before issuing a final IC1 candidate ID. Until then, candidate IDs and measured costs remain null; this PR does not add or rewrite Q proposals, candidate JSONL, curve identities or isogeny-route records. For Koblitz work preserve the repository's m83 fidelity gate before making ECC2K-130 claims; m31 or N13 controls cannot discharge that gate.

Scope and evidence

This is a literature-derived research plan dated 2026-10-21, requested as ideas for the crypto and cryptanalysis repositories. It does not introduce a solver, run an experiment, or establish a new speedup. Study-local labels DGM-01 through DGM-05 are planning labels, not candidate, curve, certificate, or run IDs. Exact experimental curve/subgroup fixtures and immutable IDs remain to be selected and frozen. Run IDs: none.

The supplied source was Duursma, Gaudry and Morain, *Speeding up the Discrete Log Computation on Curves with Automorphisms*, ASIACRYPT 1999 [S1]. We compared its mechanisms with repository notes and the primary literature listed below. The distinctions worth carrying forward are explicit orbit accounting, auxiliary-curve symmetry searches, and algebraic rejection of bad walks. Orbit folding itself is already prior art and already appears in these repos.

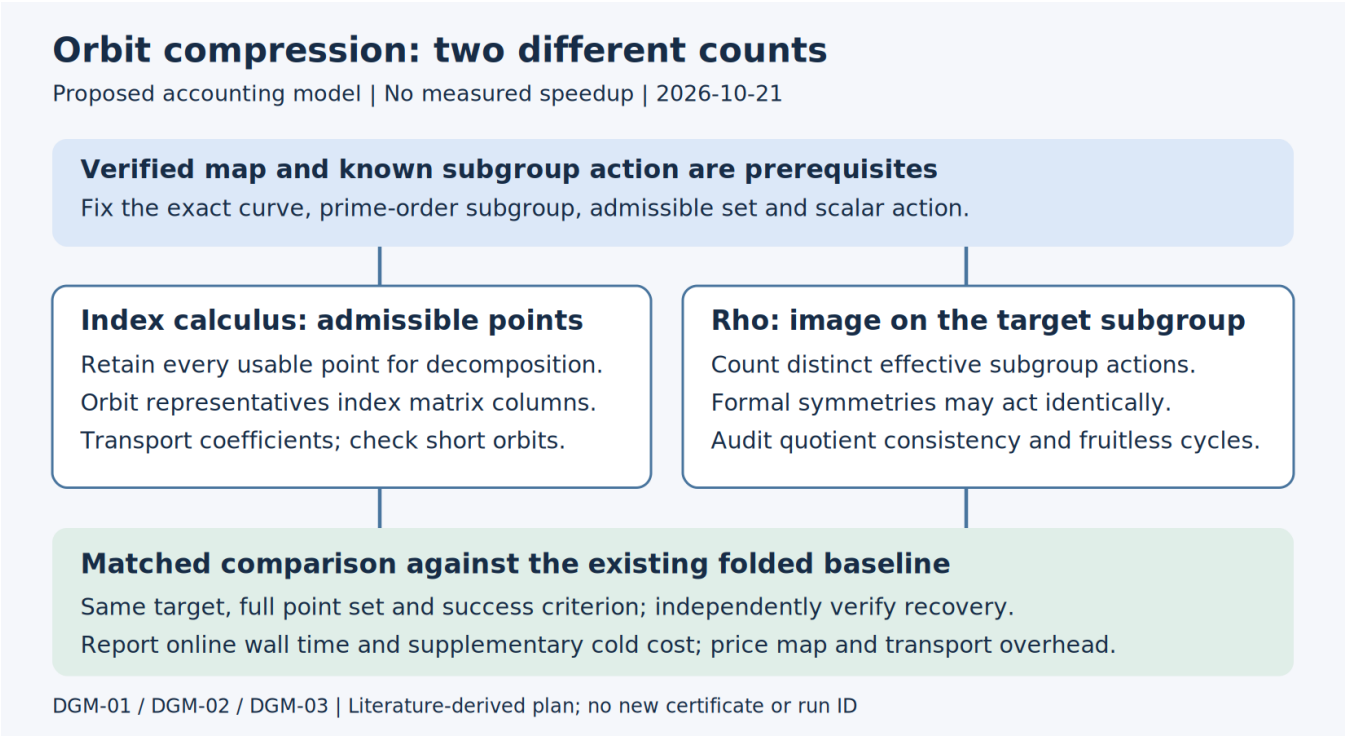


Figure 1: Proposed accounting: preserve the admissible set and distinguish matrix compression from the action on the target subgroup. No measured values.

DGM-01: preserve the factor base, compress its logarithm unknowns

Status: proposed ablation of an existing mechanism. Let $H = \langle G \rangle$ have prime order r, and let an automorphism ϕ preserve H with a known scalar action $\phi(G) = [\lambda]G$. The literature's coefficient-transport rule is

$$\log_G(\phi^i(B)) = \lambda^i \log_G(B) \pmod{r}.$$

This is a cited rule under the stated hypotheses, not a new certified identity. Before using a new map, independently verify its action and emit the repository's required evidence. A map merely acting on coordinates is insufficient.

Keep the complete, subgroup-filtered admissible set $\mathcal{B}$ for decomposition. Map each point to an orbit representative and transport its coefficient into the representative's column. Record both $|\mathcal{B}|$ and the actual number of columns; neither an orbit representative count nor a dimension is the factor base cardinality. Short orbits and stabilizers must be counted explicitly. If the original base is not stable, taking its orbit closure changes the base and must be tested as a separate treatment.

Under unchanged admissible points, decompositions remain available. An ideal factor-m reduction of columns can give a factor-m^2 reduction of a quadratic linear-algebra stage; sparsity, fill, useful rank and implementation costs can change this estimate. It is not an end-to-end speedup claim. Gaudry [S2] already develops this factor-base idea; novelty must be measured relative to the existing folded implementation, not a deliberately weakened unfurled baseline.

Acceptance: freeze the same target stream, point set and relation witnesses; independently replay unfolded and folded equations modulo r; require identical accepted decompositions and successful known-answer recovery. Compare

current folding with any proposed extension, reporting matrix rank, nonzeros, memory, transport cost and complete online wall time. Retain failed and dependent rows.

DGM-02: search the auxiliary curve for hidden symmetries

Status: proposed search, with a historical positive example. DGM's Sakai-Sakurai example uses a coordinate translation $x \mapsto x + 1$ with a polynomial adjustment of y [S1, section 4.3, Table 1]. This motivates testing affine and fractional-linear coordinate transformations on a concrete auxiliary curve or Jacobian, beyond the obvious negation and Frobenius candidates.

A candidate record needs the defining curve, coefficient field, exact map, inverse and domain, proof that the defining equation is preserved, extension across exceptional points, and the induced action on the relevant Jacobian. Then verify that the target prime-order component is stable and determine its scalar action without assuming the unknown target logarithm. Map discovery, evaluation and subgroup transport all have costs.

Separate two quantities: the orbits of admissible prime divisors used by index calculus, and the image of the automorphism group on the target subgroup used by rho. Several formal symmetries can induce the same subgroup action. A large coordinate group does not automatically enlarge rho's useful quotient.

Acceptance: produce a native replayable map/action certificate, enumerate orbit lengths on an exact small fixture, and compare the induced subgroup action with existing negation/Frobenius. Record bounded searches with no additional action as bounded negatives, not nonexistence proofs. No unverified isogeny or transfer edge is added to the curve registry.

DGM-03: audit multiplicative walks before timing them

Status: proposed correctness and mixing audit. DGM motivates inexpensive updates such as $R \mapsto R + \phi^j(R)$. Its order-four setting also illustrates the identity $(1 + \phi)(1 - \phi) = 2$ when $\phi^2 = -1$ [S1]. Before benchmarking, enumerate short update products in the endomorphism ring and look for products equal to the identity, an orbit symmetry, or another small exceptional action. A cheap step can create fruitless cycles or strong dependencies.

A usable quotient walk needs a consistent successor for every representative, an invariant partition and distinguished-point predicate, and correct transport of both coefficients in $R = aG + bQ$. Record cycles, deterministic escape actions, restart histories, and useful collisions between independent starts. Check that collision denominators are invertible before recovering a logarithm. Short-product screening is necessary evidence, not a proof of random mixing.

Modern negation-map analyses [S3, S4] must be incorporated before adopting a 1999 walk recipe. Compare the present working walk, a deliberately enabled multiplicative variant, and an audited variant on matched synthetic targets. Report failure rates and online time to independently verified recovery; additions per step alone are insufficient.

DGM-04: keep P-192/P-224 exclusions narrow and reproducible

Status: prior assessment awaiting a repository-native reproduction. The preceding paper review screened the standard NIST P-192 and P-224 parameters [S7, S8]: their j values are neither 0 nor 1728; prime-field Frobenius acts trivially on their rational points. For an ordinary curve, the trace test $t = p + 1 - r$, $\Delta = t^2 - 4p$ distinguishes whether the endomorphism algebra is $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$. The prior assessment found neither case for these two curves. This PR provides no native certificate and does not promote that assessment to an independently verified repository result.

Reproduce from the exact field, model, order and generator using native integer arithmetic, check ordinarity and the square tests, and emit the required proof or identity evidence. The intended exclusion concerns same-$\mathbb{F}_p$ isogenies to the special $j = 1728$ or $j = 0$ ordinary classes. It does not exclude all endomorphisms, extension-field routes, higher-dimensional transfers, or generic negation. An exceptional curve reached in a drawing is not a useful DLP route until its map and subgroup transport are verified.

DGM-05: separate lifting from auxiliary-input attacks

Status: literature-gated proposal. DGM discusses characteristic-zero lifting and Xedni. The primary Xedni proposal [S5] and subsequent analysis [S6] were consulted at abstract level only in this review. The analysis describes

severe lifting/dependence obstacles and asymptotic failure of the original method. Those abstracts do not justify closing every possible lifting variant.

Read the full papers before designing a run. State precisely which obstruction a variant changes, how dependent lifts are produced, the height/rank costs, and how a relation yields the target logarithm without an oracle. Treat any rank-one/Wieferich mechanism as a separately sourced hypothesis, not an implemented continuation of Xedni. Keep this distinct from Cheon's attack: auxiliary powers of the unknown scalar change the input problem. Any such assumption must appear explicitly in both the candidate and the comparator.

Acceptance: a complete algorithm and input contract, a worked public known-answer example, and a native independent checker before timing or funding. A literature gap is an open check, not evidence of a working attack.

Measurement contract for future work

The primary comparison is the matched single-target ONLINE wall ratio $T_{\rho,\text{online}}/T_{\text{candidate,online}}$, on the same target, subgroup, hardware and success criterion. Report reusable setup separately and supply full cold cost as a supplementary view. Price canonicalization, map evaluation, relation search, rank recovery, target descent and final verification in their appropriate intervals.

Freeze public synthetic/known-answer fixtures, code revision, seeds, resource caps, raw logs and verifier output. Do not average away failures or present batch amortization as a one-target win. Tiny fixtures establish correctness only; Koblitz small-field diagnostics do not settle ECC2K-130 behavior. No wall times, samples, confidence intervals or newly verified algebraic certificates exist for the proposals in this report.

Primary sources and reading status

- **S1 (full supplied paper):** Duursma, Gaudry, Morain (1999), Speeding up the Discrete Log Computation on Curves with Automorphisms, especially sections 2.3, 3, 4.2-4.3 and 6.
- **S2 (full paper):** Gaudry (2000), An Algorithm for Solving the Discrete Log Problem on Hyperelliptic Curves, section 4.2. Read through a mirror of the primary paper.
- **S3 (full paper):** Bos, Kleinjung, Lenstra, On the Use of the Negation Map in the Pollard Rho Method.
- **S4 (abstract/metadata):** Bernstein, Lange, Schwabe (2011), On the correct use of the negation map in the Pollard rho method. Full implementation details remain a pre-experiment reading gate.
- **S5 (abstract only):** Silverman, The Xedni Calculus and the Elliptic Curve Discrete Logarithm Problem.
- **S6 (abstract only):** Analysis of the Xedni Calculus Attack.
- **S7 (parameter sections):** NIST, FIPS 186-4, historical P-192 parameters. This withdrawn edition is cited for the historical curve, not current deployment guidance.
- **S8 (parameter sections):** NIST, SP 800-186, P-224 parameters.

Activation and graph record

The companion crypto backlog card `XP-20261021-automorphism-ideas` is blocked and unfunded. This repository's activation checklist is:

1. Reconcile DGM-01 with the exact current folding implementation and select registered known-answer fixtures; pin the full admissible point set.
2. Implement a native independent equation/action verifier and the ablation interface; freeze the protocol, code revision, seeds and resume behavior.
3. Verify quotient and scalar transport on exact small fixtures before any performance run. Keep each of DGM-02 through DGM-05 blocked until its own map, correctness or literature gate is satisfied.
4. Run and retain the requested matched controls only in a later experiment task; report failures, raw outputs and whole online wall intervals.

Checked the catalog's `curves.yaml` and the figure inventory under `experiments/volcano-ic/figures/` (census, ECDLP and tau views). No new registered curve, certified edge, measured cost or changed historical conclusion follows from this plan, so no canonical graph data, rendered graph, or paper PDF is revised. There is no quantitative graph because there are no new observations to plot.

orbit-compression.svg is the editable diagram source. Makefile builds the PDF from this README. VALIDATION .md records the document checks and native gates.